

INDIAN SCHOOL OF BUSINESS

AMPBA 2026 Summer, Term 5

LLMs & Generative AI

Group Assignment Report

Submitted by Group 24

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Part A: Industry Applications of Computer Vision Tasks

1. Image Classification

Image classification assigns a single categorical label to an entire image, paired with a statistical confidence score. It is primarily used for automated categorization, triaging, and quality assurance.

Domain	Input Data	Expected Output	Deployed at	Reference
Healthcare	Chest X-rays (DICOM) from hospital PACS	Disease label: Normal / Pneumonia / COVID-19 + confidence score	Google Health / CheXNet Stanford	Rajpurkar et al., CheXNet (arXiv:1711.05225)
Retail	Product photos uploaded by marketplace sellers	Taxonomy label (e.g., Men > Footwear > Sneakers) for auto-cataloging	Amazon Rekognition Custom Labels	AWS Blog
Agriculture	Smartphone leaf photos captured by farmers	Crop disease class (Leaf Blight, Healthy) + treatment advice	Plantix / PEAT GmbH (10M+ farmers)	Mohanty et al., Frontiers in Plant Science 2016
Manufacturing	Line-scan camera images of PCBs and castings	Pass/Fail/Defect-type label triggering auto-rejection	Foxconn / Samsung assembly lines	Cha et al., Computer-Aided Civil & Infra Engg 2017

2. Image Similarity Computation

Image similarity maps images into a low-dimensional embedding space to calculate visual proximity, powering reverse-image search, discovery engines, and diagnostic verification.

Domain	Input Data	Expected Output	Deployed at	Reference
E-Commerce	User photo of product + indexed catalog of millions of images	Ranked visually similar products; 'shop the look' results	Pinterest Lens (150M+ searches/mo)	Zhai et al., Visual Discovery at Pinterest (arXiv:1702.04680)
IP Protection	New image uploads vs. copyrighted stock photo database	Similarity score + matched source; flag unlicensed use	Getty Images / Shutterstock	TinEye Reverse Image Search
Healthcare	New patient CT/MRI vs. annotated historical scan database	Top-K similar historical cases for differential diagnosis	IBM Watson Health (Merative)	Müller et al., Int. J. Medical Informatics 2004
Fashion	Cart item image vs. full catalog embeddings	Complementary outfit items ranked by visual style similarity	ASOS / Zalando	Vittayakorn et al., Runway to Realway (IEEE WACV 2015)

3. Object Detection

Object detection localizes multiple objects within a frame by drawing precise bounding boxes around them while simultaneously identifying their class labels.

Domain	Input Data	Expected Output	Deployed at	Reference
Retail Cashierless	Overhead + shelf RGB camera feeds inside store in real time	Detected products + customer pick/return events for billing	Amazon Just Walk Out Technology	Amazon Just Walk Out

Autonomous Driving	Multi-camera + LiDAR streams at 30+ FPS from road	Bounding boxes: pedestrians, vehicles, signs as inputs for path planner	Waymo / Tesla Autopilot	Geiger et al., KITTI Benchmark (CVPR 2012)
Healthcare Pathology	Whole-slide H&E biopsy images at 40x magnification	Bounding boxes on tumor regions + malignancy probability score	PathAI / Google DeepMind	Bulten et al., Lancet Oncology 2020
Agriculture	Drone RGB aerial imagery of crop fields during season	Detected pests / ripe fruit clusters for precision spraying	John Deere - Blue River Technology	Lottes et al., ICRA 2017

4. Image Segmentation

Segmentation moves beyond bounding boxes to classify individual pixels, yielding fine-grained spatial masks of target boundaries.

Domain	Input Data	Expected Output	Deployed at	Reference
Healthcare Surgery	MRI/CT scan volumes from hospital DICOM pre-surgery	Pixel masks of organs and tumors for volume calculation and planning	Siemens Healthineers AI-Rad Companion	Ronneberger et al., U-Net (MICCAI 2015)
Autonomous Driving	Front-camera video stream at 30 FPS in varied conditions	Dense pixel labels: road, pedestrian, vehicle, sky per frame	Waymo / Mobileye / Baidu Apollo	Cordts et al., Cityscapes Dataset (CVPR 2016)
Remote Sensing	Sentinel-2 multi-spectral satellite imagery at 10m resolution	Land-use map: forest, urban, water, cropland pixels	Planet Labs / ESA	Helber et al., EuroSAT (IEEE JSTARS 2019)
Manufacturing	Line-scan images of semiconductor wafers and textiles	Instance masks outlining defect regions for root-cause analysis	Applied Materials / TSMC	Bergmann et al., MVTec AD (CVPR 2019)

5. Visual Question Answering (VQA)

VQA crosses modalities by enabling users to ask open-ended natural language questions about an image and receive contextually accurate text or spoken answers.

Domain	Input Data	Expected Output	Deployed at	Reference
Healthcare Radiology	X-ray/CT image + question: <i>"Is there pleural effusion?"</i>	NL answer: <i>"Yes, mild bilateral effusion in lower lobes"</i>	Microsoft Nuance radiology reporting	Lin et al., Medical VQA Survey (AI in Medicine 2023)
Accessibility	Smartphone camera frame + voice query about label/text	Spoken answer with medicine name, dose, instructions	Microsoft Seeing AI global deployment	Microsoft Seeing AI
E-Commerce	Product image + customer question about features	Answer derived from visual cues + product specs inline	Amazon / Flipkart product Q&A	Gao et al., TextVQA (CVPR 2020)
Industrial Inspection	Equipment photo + query: <i>"Is there corrosion on the flange?"</i>	<i>"Yes, moderate rust on upper-left flange. Maintenance needed."</i>	GE Digital / IBM Maximo	Shao et al., Technical VQA (IEEE CASE 2022)

6. Action Recognition

Action recognition analyzes sequential video frames to detect and log human movements, behavioral events, or complex physical activities.

Domain	Input Data	Expected Output	Deployed at	Reference
Sports Analytics	Multi-angle broadcast video at 25-60 FPS	Per-player action labels: Dribble, Pass, Shoot, Sprint for reporting	Stats Perform / Second Spectrum (NBA, EPL)	Giancola et al., SoccerNet (CVPR Workshop 2018)
Retail - Loss Prevention	CCTV video from store entry and exit zones	Alert: " <i>Concealment event</i> " with timestamp + camera ID	Verint / Checkpoint Systems	Arroyo et al., Expert Systems with Applications 2015
Elderly Care	Depth/RGB footage from care-home rooms 24/7	Alert: " <i>Fall Detected</i> " within seconds triggering caregiver notification	Alert-1 / Bay Alarm Medical	Núñez-Marcos et al., Wireless Comm & Mobile Computing 2017
Workplace Safety	Live CCTV on construction sites and factory floors	Labels: " <i>Helmet Removed</i> ", " <i>Worker in Restricted Zone</i> " triggering a real-time alert	Intenseye / Protex AI	Nath et al., Automation in Construction 2020

7. Multi-Object Tracking (MOT)

MOT tracks objects of interest continuously across frames while keeping a persistent identity for each item, tracing trajectories over time.

Domain	Input Data	Expected Output	Deployed at	Reference
Autonomous Driving	Camera + LiDAR streams from self-driving vehicles at 30 FPS	Unique IDs for all pedestrians and vehicles + predicted trajectories	Waymo / Tesla / Mobileye	Weng et al., AB3DMOT (IROS 2020)
Retail Analytics	Overhead fisheye camera feeds in supermarkets and malls	Anonymized customer trajectories, heat maps, queue lengths	RetailNext / Sensormatic (Johnson Controls)	Leal-Taixé et al., MOTChallenge 2015
Sports Player Tracking	8-12 synchronized broadcast cameras at 50 FPS	XY position tracks for all 22 players + ball every frame	FIFA / NBA via ChyronHego and Genius Sports	Held et al., Learning to Track at 100 FPS (ECCV 2016)
Urban Traffic Mgmt	Fixed CCTV at intersections: continuous 24/7 feeds	Per-lane count, speed, incident detection for adaptive traffic lights	Conduent / Yunex Traffic smart city	Bochinski et al., High-Speed Tracking-by-Detection (AVSS 2017)

8. Image and Video Generation

Generative AI uses diffusion models and GANs to synthesize realistic images, high-fidelity videos, or synthetic training data from textual or structural parameters.

Domain	Input Data	Expected Output	Deployed at	Reference
Marketing & Advertising	Text prompt + brand style guide / color palette	High-res product images for ad campaigns - replaces photo shoots	Coca-Cola / Heinz (DALL-E 3); Adobe Firefly	Rombach et al., Latent Diffusion Models (CVPR 2022)
Gaming - Asset Creation	Text or concept art descriptions + art-style references	Game-ready 2D/3D assets, textures, environments at scale	EA / Ubisoft use Midjourney v6; NVIDIA Canvas	Midjourney v6; NVIDIA Canvas
Film & Media Production	Script scene descriptions or storyboard frames as prompts	Up to 1-min HD video clips for pre-visualization and B-roll	OpenAI Sora / Runway Gen-3	OpenAI Sora: Video Generation Models as World Simulators
Healthcare Synthetic Data	Small real medical image dataset + augmentation parameters	Synthetic labeled MRI/CT/pathology images for AI model training	GE Healthcare / Siemens Healthineers	Frid-Adar et al., GAN-based Medical Image Augmentation (CMIG 2018)

PART B – Image Classification with Pretrained models (ResNet18, DenseNet121, VGG19)

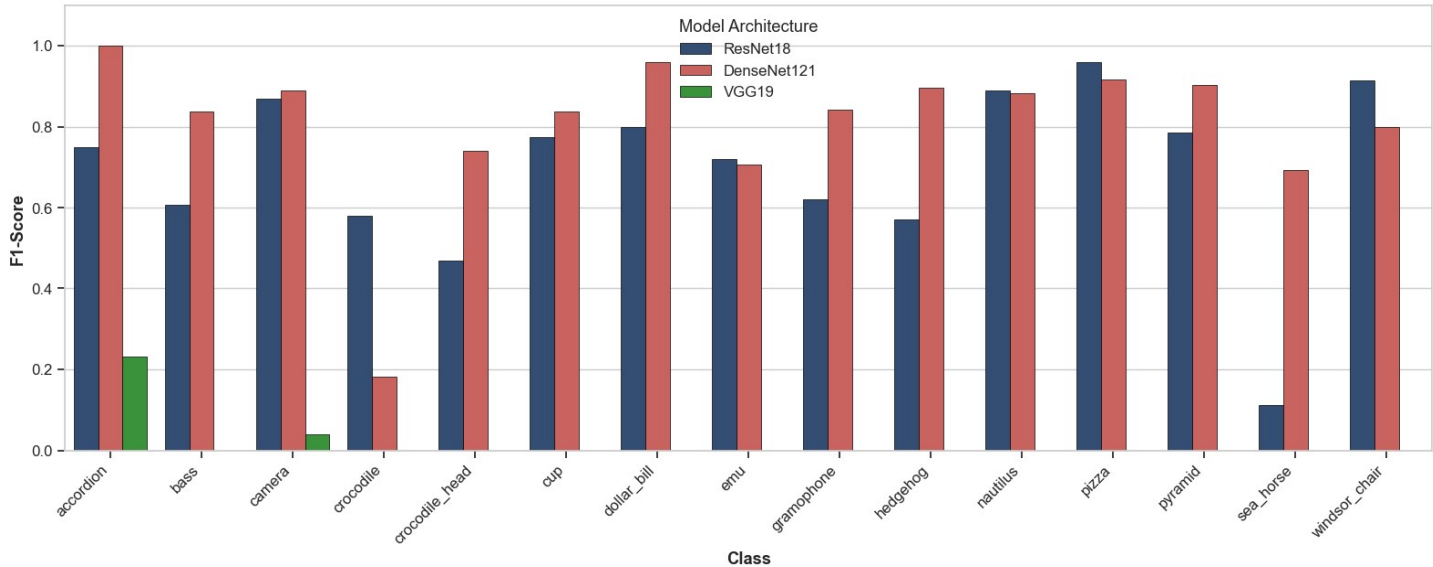
Q1 - Pretrained CNNs (ResNet18, DenseNet121, VGG19) on 15-Class Dataset

- **Objective:** Train 15-class image classifiers using pretrained ResNet18, DenseNet121, and VGG19; report per-class precision and recall on the defined test split.
- **Dataset & Split:** 805 images across 15 categories; per-class split rule → first 40 images for training, remainder for testing; observed counts: Train = 600, Test = 205.
- **Modeling Setup:** Pretrained backbones with final layer replaced for 15 outputs; input resized to 224×224; augment with RandomHorizontalFlip; normalize with ImageNet mean/std.
- **Training Details:** CrossEntropyLoss, Adam optimizer (lr=1e-3), 3 epochs, batch size 32.
- **Evaluation:** Predictions via argmax; metrics per class using precision, recall, F1, and support

On running the setup, we observed the following evaluation results on test set.

class	RESNET				DENSENET				VGGNET			
	precision	recall	f1-score	support	precision	recall	f1-score	support	precision	recall	f1-score	support
accordion	1.00	0.60	0.75	15.00	1.00	1.00	1.00	15.00	0.13	1.00	0.23	15.00
bass	0.44	1.00	0.61	14.00	0.76	0.93	0.84	14.00	0.00	0.00	0.00	14.00
camera	0.77	1.00	0.87	10.00	1.00	0.80	0.89	10.00	0.02	0.20	0.04	10.00
crocodile	0.43	0.90	0.58	10.00	1.00	0.10	0.18	10.00	0.00	0.00	0.00	10.00
rocodile_head	0.67	0.36	0.47	11.00	0.63	0.91	0.74	11.00	0.00	0.00	0.00	11.00
cup	0.86	0.71	0.77	17.00	0.93	0.76	0.84	17.00	0.00	0.00	0.00	17.00
dollar_bill	0.67	1.00	0.80	12.00	0.92	1.00	0.96	12.00	0.00	0.00	0.00	12.00
emu	0.75	0.69	0.72	13.00	0.57	0.92	0.71	13.00	0.00	0.00	0.00	13.00
gramophone	0.50	0.82	0.62	11.00	1.00	0.73	0.84	11.00	0.00	0.00	0.00	11.00
hedgehog	0.86	0.43	0.57	14.00	0.87	0.93	0.90	14.00	0.00	0.00	0.00	14.00
nautilus	1.00	0.80	0.89	15.00	0.79	1.00	0.88	15.00	0.00	0.00	0.00	15.00
pizza	1.00	0.92	0.96	13.00	1.00	0.85	0.92	13.00	0.00	0.00	0.00	13.00
pyramid	1.00	0.65	0.79	17.00	1.00	0.82	0.90	17.00	0.00	0.00	0.00	17.00
sea_horse	1.00	0.06	0.11	17.00	1.00	0.53	0.69	17.00	0.00	0.00	0.00	17.00
windsor_chair	0.84	1.00	0.91	16.00	0.67	1.00	0.80	16.00	0.00	0.00	0.00	16.00

F1-Score per Class across Architectures

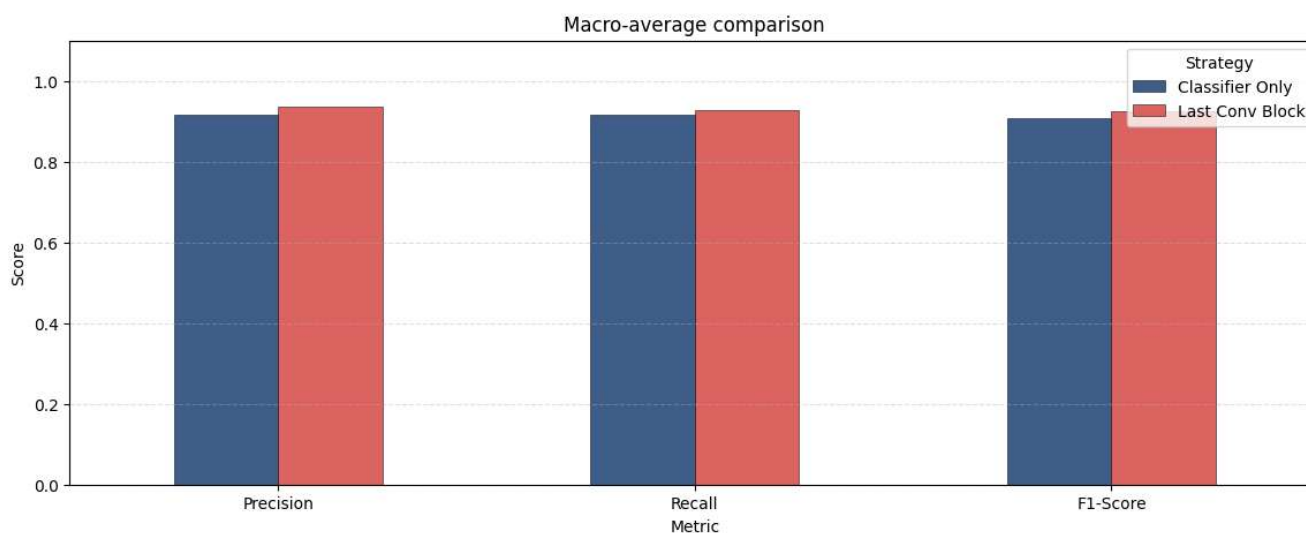


Q2 - Finetuning VGG19 model for the same 15 class dataset

We further finetuned VGG19 model with two approaches.

- 1) Finetuning only the classifier head of VGG19 and Freeze all feature layers.
- 2) Finetuning layers from last convolutional block. VGG19 features have 37 layers. Last block starts around layer 24.

class	Finetuning only Classifier				Finetuning from last convolutional block			
	precision	recall	f1-score	support	precision	recall	f1-score	support
accordion	0.79	1.00	0.88	15.00	0.94	1.00	0.97	15.00
bass	1.00	1.00	1.00	14.00	0.82	1.00	0.90	14.00
camera	1.00	1.00	1.00	10.00	1.00	1.00	1.00	10.00
crocodile	0.55	0.60	0.57	10.00	0.86	0.60	0.71	10.00
crocodile_head	0.67	0.91	0.77	11.00	0.60	0.82	0.69	11.00
cup	1.00	1.00	1.00	17.00	1.00	1.00	1.00	17.00
dollar_bill	1.00	1.00	1.00	12.00	0.92	1.00	0.96	12.00
emu	1.00	1.00	1.00	13.00	1.00	1.00	1.00	13.00
gramophone	1.00	1.00	1.00	11.00	1.00	0.91	0.95	11.00
hedgehog	1.00	0.93	0.96	14.00	1.00	1.00	1.00	14.00
nautilus	0.94	1.00	0.97	15.00	1.00	1.00	1.00	15.00
pizza	0.93	1.00	0.96	13.00	1.00	1.00	1.00	13.00
pyramid	1.00	0.71	0.83	17.00	1.00	0.82	0.90	17.00
sea_horse	1.00	0.59	0.74	17.00	1.00	0.76	0.87	17.00
windsor_chair	0.89	1.00	0.94	16.00	0.89	1.00	0.94	16.00



Post-finetuning, the overall correctness of the model has significantly improved. We can also observe that classifier-only fine-tuning is safer for classes with clear features, while deeper fine-tuning helps recover recall and balance performance for visually complex or overlapping categories.

Q3 - Using CLIP-ViT-B/16 model for evaluating zero-shot per-class classification accuracy.

We evaluated zero-shot classification using the CLIP-ViT-B/16 model across 15 classes, measuring precision, recall, and F1 scores without any task-specific fine-tuning.

class	precision	recall	f1-score	support
accordion	1.00	1.00	1.00	55.00
bass	1.00	1.00	1.00	54.00
camera	1.00	1.00	1.00	50.00
crocodile	0.76	0.74	0.75	50.00
crocodile_head	0.75	0.76	0.76	51.00
cup	1.00	1.00	1.00	57.00
dollar_bill	1.00	1.00	1.00	52.00
emu	1.00	1.00	1.00	53.00
gramophone	0.98	1.00	0.99	51.00
hedgehog	1.00	1.00	1.00	54.00
nautilus	1.00	0.98	0.99	55.00
pizza	1.00	1.00	1.00	53.00
pyramid	1.00	1.00	1.00	57.00
sea_horse	0.98	0.98	0.98	57.00
windsor_chair	1.00	1.00	1.00	56.00

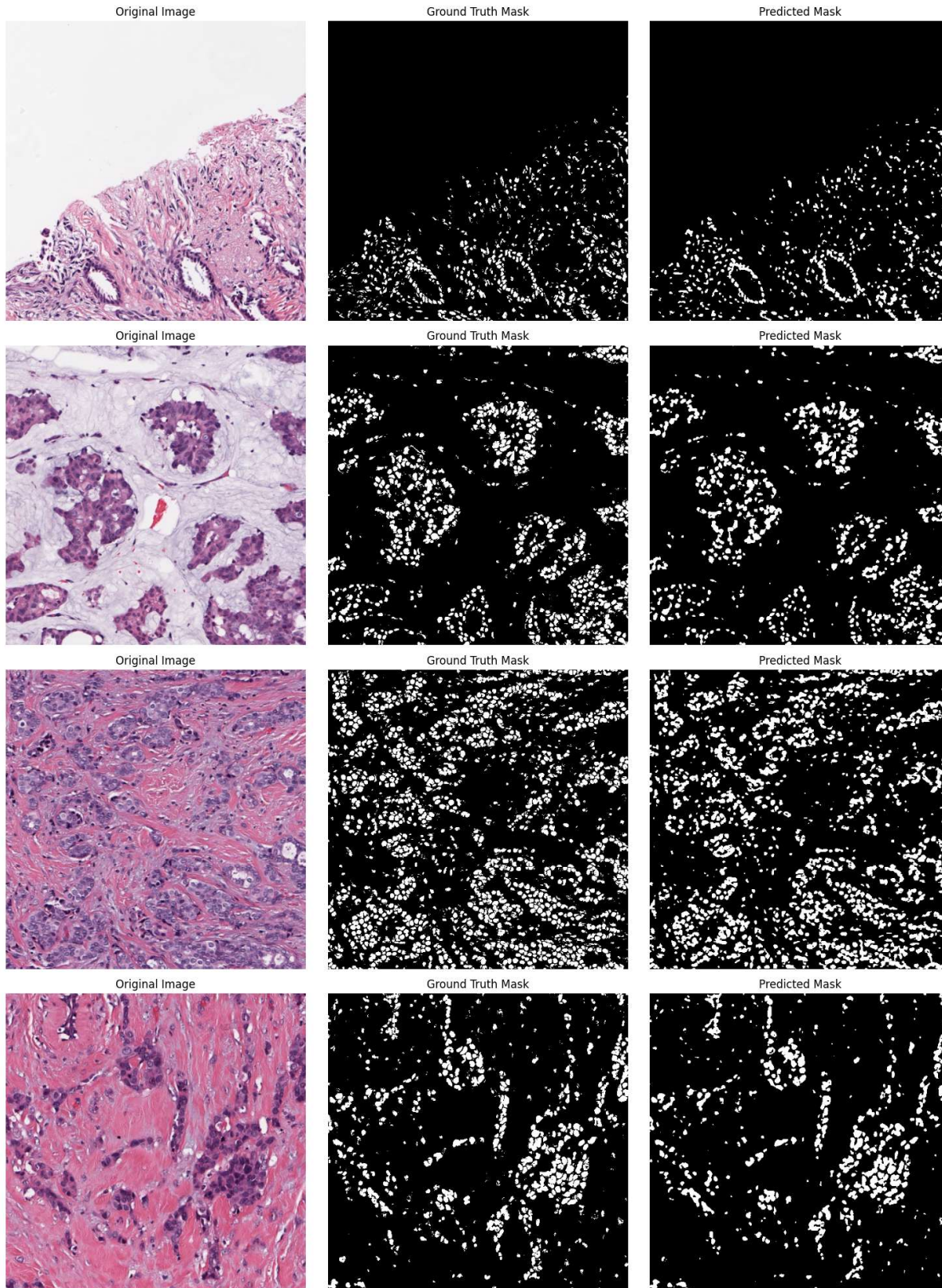
CLIP delivered near-perfect accuracy for most categories and consistent performance overall, outperforming fine-tuned CNN baselines in stability, with only minor gaps in fine-grained classes like *crocodile* and *crocodile_head*.

PART C – Finetune UNet model with ResNet-18 backbone for segmentation.

Q4 - Finetune UNet model for segmentation on 24 test images

We finetuned UNet model with 20 epochs. On evaluating the Test Set Metrics for Nucleus Class, we get the Precision: 0.8315 and Recall: 0.7876.

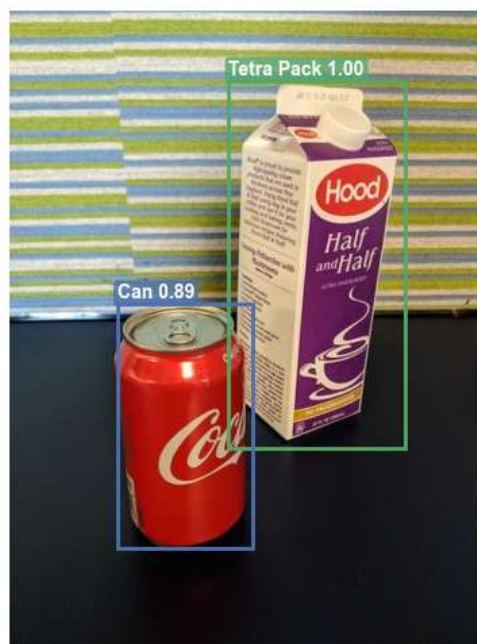
Below is the comparison of original image, Ground truth mask, and the predicted mask.



PART D – Finetune a Faster RCNN model for object detection.

Q5 - Trained a Faster R-CNN detector on 100 images (with a new classifier head for 4 object classes plus background) and evaluated on 28 test images at IoU ≥ 0.5 , reporting per-class precision and recall.

Following is the prediction on samples, label alignment and precision and recall for each class.



- Can — Precision 0.4138, Recall 1.0000
- Tetra Pack — Precision 0.4444, Recall 1.0000
- Milk Bottle — Precision 0.4138, Recall 1.0000
- Water Bottle — Precision 0.4118, Recall 1.0000

The model achieved perfect recall across all classes (it detected every ground-truth object) but moderate precision (~0.41–0.44), meaning many false positives; this suggests the detector is highly sensitive but needs tighter confidence thresholds.